INTRODUCTION TO DATA MANAGEMENT PROJECT REPORT

(Project Semester January-April 2025)

PUBLIC_LIBRARIES

Submitted by

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Registration No.: 12307183

Programme: BTech-CSE

Section: K23GW

Course Code: INT-217

Under the Guidance of

Baljinder Kaur (27952)

Discipline of CSE/IT

Lovely School of Computer Science Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that **Roshani Kumari** bearing Registration no. **12307183** has completed **INT-217** project titled, **“Public_Libraries”** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Baljinder Kaur

Professor

School of Computer Science Engineering

Lovely Professional University Phagwara, Punjab.

Date: 10^{th} April, 2025

DECLARATION

I, Roshani Kumari, student of BTech under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 10^{th} April 2025 Signature

Registration No. 12307183 Roshani Kumari

Acknowledgment

I would like to express my deepest gratitude to Prof. Baljinder Kaur, for her exceptional mentorship and unwavering support throughout the duration of this project. Her vast knowledge in the fields of data science and machine learning, combined with her patient and thoughtful guidance, played a pivotal role in the successful completion of this work. Her insightful suggestions and feedback consistently challenged me to think critically and improve the quality of my research. I am also grateful for the learning environment she fostered, which encouraged exploration and innovation.

In addition, I sincerely thank my peers and classmates for their helpful discussions, encouragement, and collaborative spirit during this project. Their input provided fresh perspectives that contributed meaningfully to the final outcome. I am also thankful to the open-source community for providing the tools, libraries, and resources that made the implementation of this project possible. Lastly, I acknowledge the dataset contributors for making this analysis feasible.

1.Introduction

Libraries have always been essential institutions that promote education, knowledge sharing, and cultural development. With the rise of digital transformation, tracking library usage and performance over time is vital for maintaining relevance and meeting community needs.

This project aims to build an interactive Excel dashboard using public library data to visualize key insights such as:

- Year-wise trends in library visits
- Distribution of library types across counties
- Total item circulations by location

The dashboard is designed to be dynamic and user-friendly, enabling viewers to filter and explore the data using slicers, without writing a single formula. The objective is to present meaningful information through effective data visualization tools available in Microsoft Excel.

2. Source of Dataset

The dataset used in this project is publicly available and sourced from a government or open data portal, specifically related to Public Library Statistics. It contains information about library visits, circulations, library types, locations, fiscal years, and more, across multiple counties like Hartford, Fairfield, and New Haven. Dataset Features:

- Covers multiple fiscal years
- Includes records from multiple counties (e.g., Fairfield, Hartford, New Haven)
- Offers numerical and categorical fields ideal for analysis

Dataset Link: https://catalog.data.gov/dataset/public-libraries-b1aaf

3.Dataset Preprocessing

The dataset was initially cleaned in Excel by:

- Removing empty rows and irrelevant columns
- Standardizing column headers
- Ensuring correct data types (e.g., numeric for visit counts and circulations)
- Formatting fiscal year and county fields consistently No formula-based preprocessing was used.

All preparation steps were done manually or through built-in Excel tools such as filters, sorting, and formatting.

4. Analysis on Dataset(For each Objective)

Objective 1: Public Library Visits vs Fiscal Years i. General Description:

Analyzes how public engagement with libraries (in terms of visits) has evolved over different fiscal years.

ii. Specific Requirements:

Display total visits across fiscal years and identify notable changes or trends

iii. Analysis Results:

The data reveals fluctuations in library visit counts, possibly influenced by social factors, public policy, or technological trends. Some fiscal years recorded spikes in engagement, while others showed noticeable dips.

iv. Visualization:

A Line Chart was used to represent this objective, clearly showing peaks and troughs in visit numbers across time.

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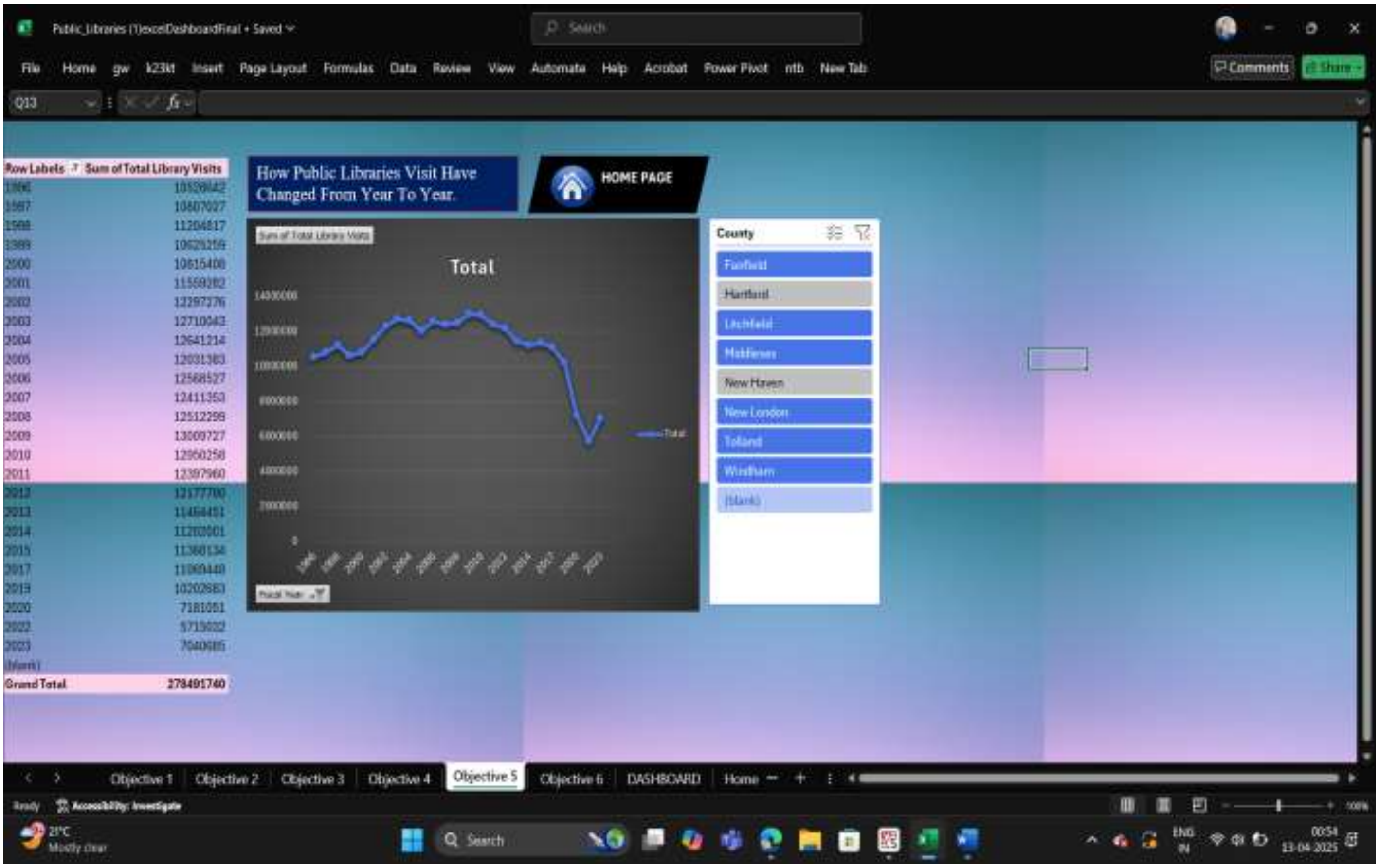

Public_Libraries (1)excelDashboardFinal • Saved
File Home gw k23kt Insert Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot ntb New Tab
Comments
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Sum of Total Library Visits
Grand Total
278491740
How Public Libraries Visit Have Changed From Year To Year.
HOME PAGE
Total
County
Fairfield
Hartford
Litchfield
Middlesex
New Haven
New London
Tolland
Windham
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Objective 1
Objective 2
Objective 3
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Objective 5
Objective 6
DASHBOARD
Home

Objective 2: Library Types Distribution

i. General Description:

Examines how libraries are distributed across counties by type or classification.

ii. Specific Requirements:

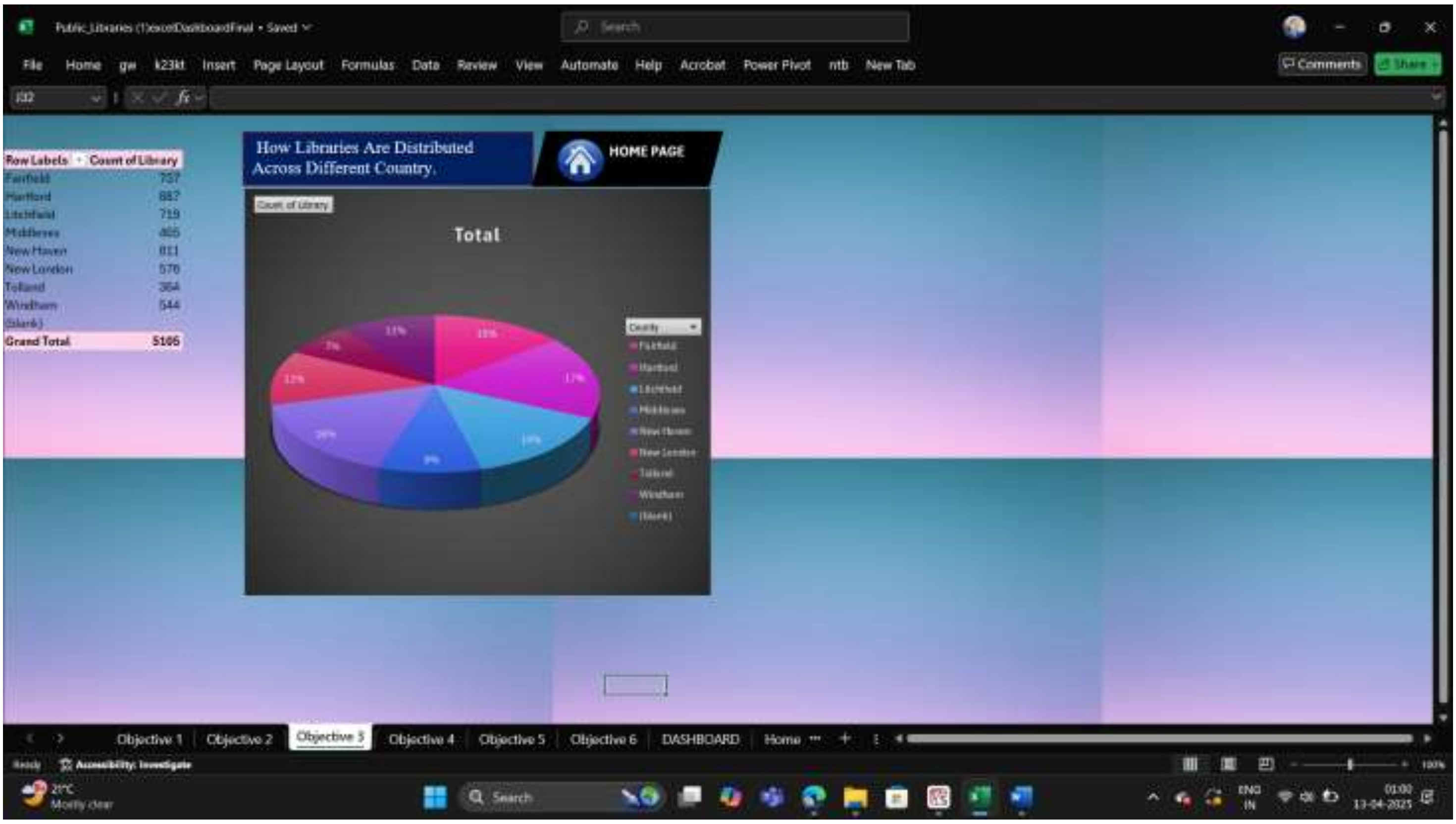
Identify regional differences in the presence of library types across major counties.

iii. Analysis Results:

Counties like Hartford and Fairfield host more libraries compared to others. The distribution suggests areas with greater population density or administrative investment in public infrastructure.

iv. Visualization:

A 3D Pie Chart was used, with each segment representing a different county. This visual makes it easy to compare library distribution at a glance.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Comments
Share
Row Labels
Count of Library
Fairfield
Hartford
Litchfield
Middlesex
New Haven
New London
Tolland
Windham
(blank)
Grand Total
How Libraries Are Distributed
Across Different Country.
HOME PAGE
Total
Objective 1
Objective 2
Objective 3
Objective 4
Objective 5
Objective 6
DASHBOARD
Home

Objective 3: Total Library Circulations by Country

i. General Description:

Determines how many library items (books, media, etc.) are circulated in each county.

ii. Specific Requirements:

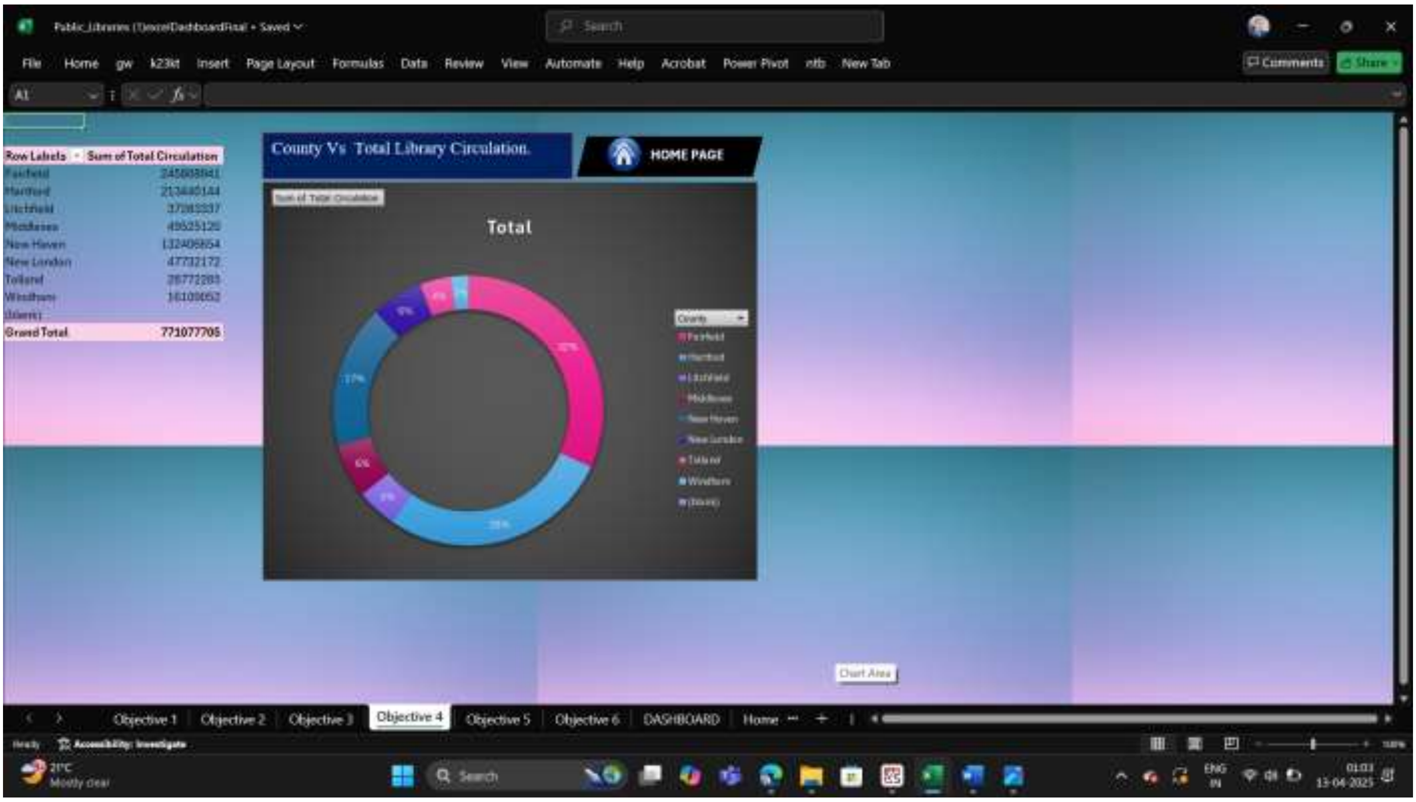
Highlight active library systems based on user engagement with physical or digital resources.

iii. Analysis Results:

Counties with larger populations and better-funded libraries tend to show higher circulation numbers. This metric gives insight into how actively libraries are being used for lending services.

iv. Visualization:

A Doughnut Chart effectively demonstrates proportional differences in item circulation across counties.



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Sum of Total Circulation
Grand Total
771077705
County Vs Total Library Circulation.
HOME PAGE
Total
Chart Area
Objective 1
Objective 2
Objective 3
Objective 4
Objective 5
Objective 6
DASHBOARD
Home

Objective 4: Operational Expenditures

i. General Description:

Compares how much various counties spend on operating their public libraries.

ii. Specific Requirements:

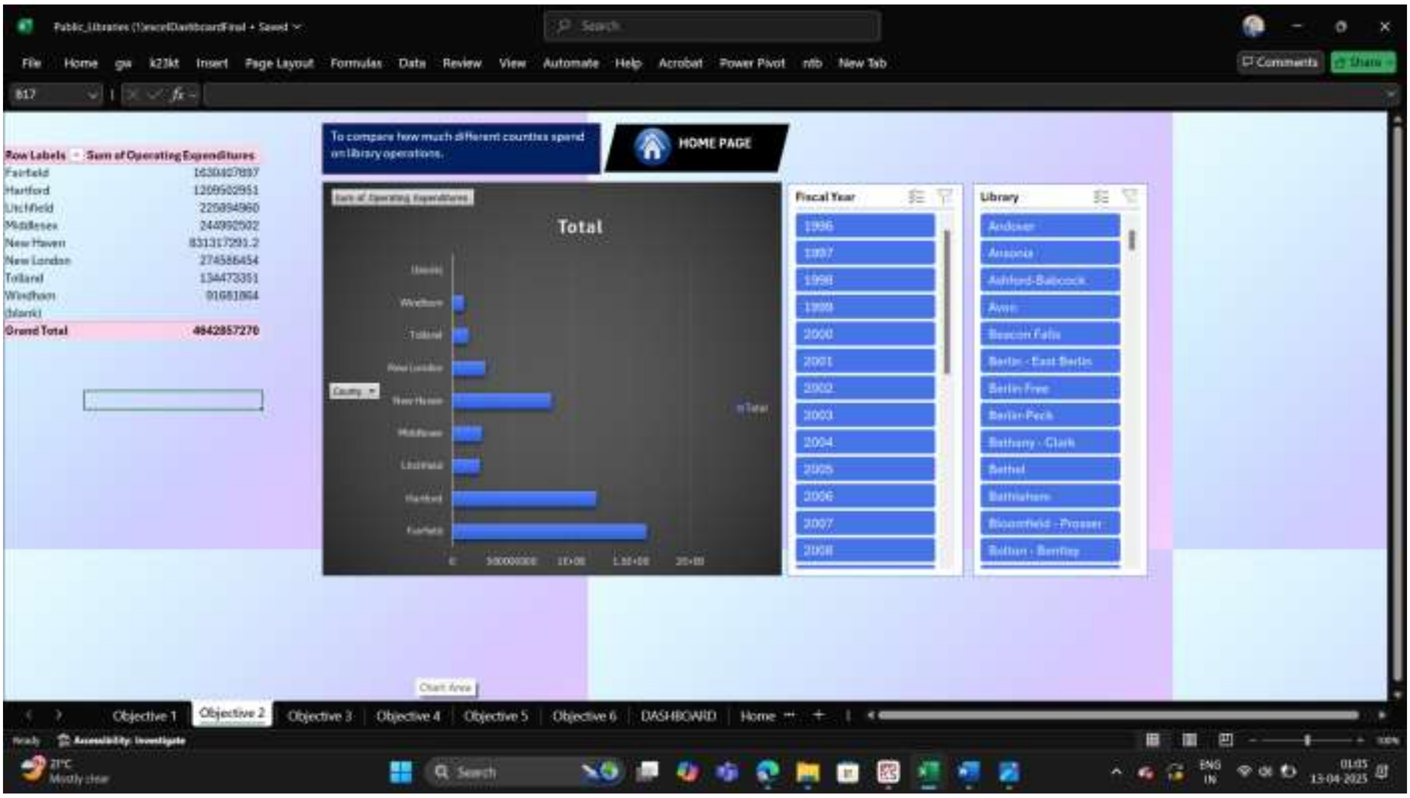
Showcase financial investments in library infrastructure and maintenance.

iii. Analysis Results:

There is a significant variation in expenditure across counties. Some counties allocate larger budgets, indicating a higher level of support for public services.

iv. Visualization:

A Horizontal Bar Chart was used for clear, side-by-side comparison of spending levels among different counties.


File Home Insert Page Layout Formulas Data Review View Automate Help Acrobat Power Pivot New Tab
Row Labels
Sum of Operating Expenditures
Fairfield 1630407897
Hartford 1209502951
Litchfield 225894960
Middlesex 244992502
New Haven 831317291.2
New London 274586454
Tolland 134473351
Windham 91681864
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Grand Total 4642857270
To compare how much different counties spend on library operations.
HOME PAGE
Total
Fiscal Year
1996
1997
1998
1999
2000
2001
2002
2003
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2007
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Library
Andover
Ansonia
Ashford-Babcock
Avon
Beacon Falls
Berlin - East Berlin
Berlin Free
Berlin-Peck
Bethany - Clark
Bethel
Bethlehem
Bloomfield - Prosser
Bolton - Bentley
Objective 1
Objective 2
Objective 3
Objective 4
Objective 5
Objective 6
DASHBOARD
Home

Objective 5: Library Collection Sizes

i. General Description:

Assesses the total collection size (books, digital resources, etc.) held by libraries.

ii.Specific Requirements:

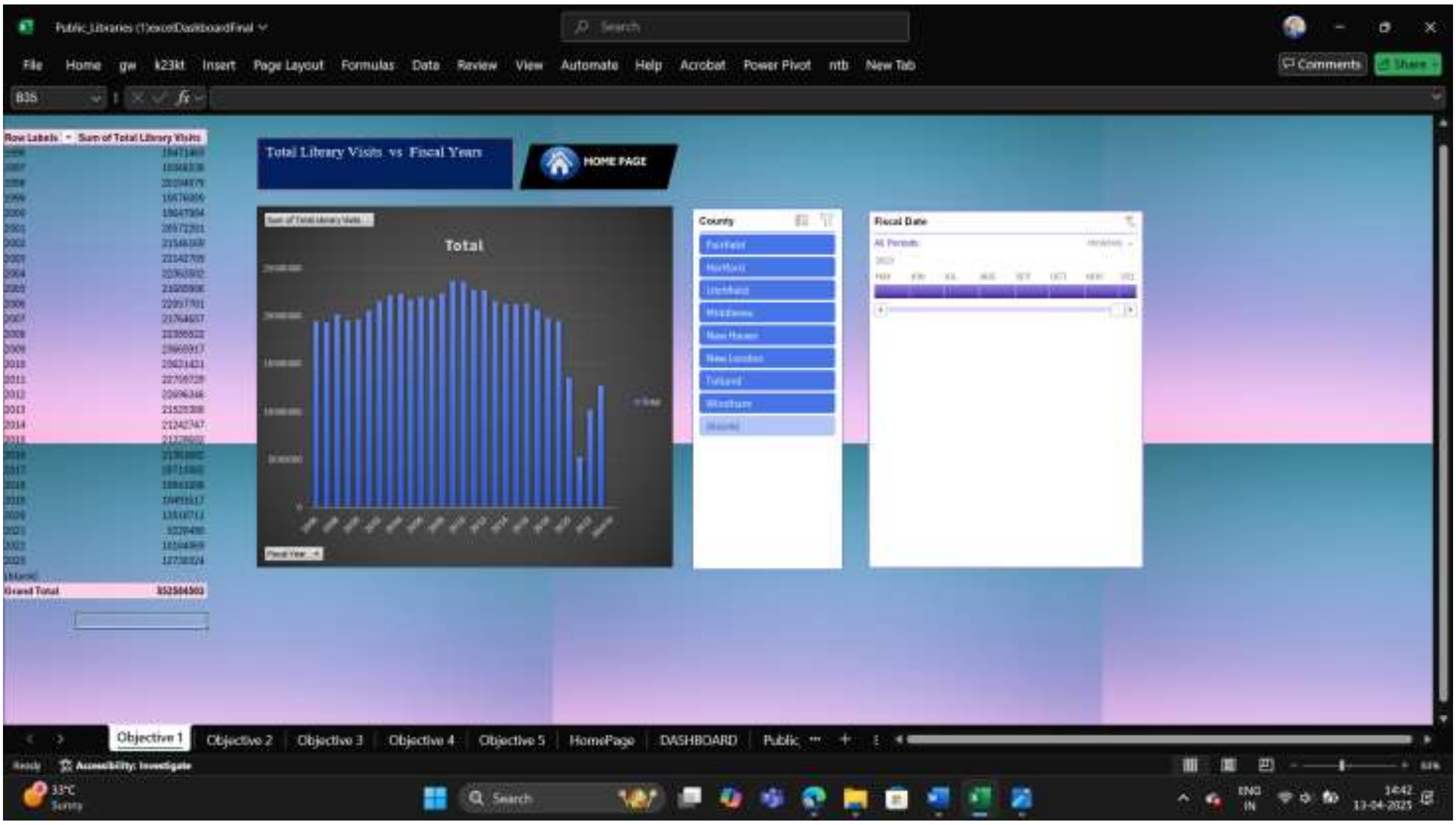
Visualize how collection sizes vary across libraries to identify resource-rich and resource-scarce institutions.

iii. Analysis Results:

While most libraries maintain an average-sized collection, a few stand out with significantly larger or smaller inventories, possibly due to their size, age, or specialization.

iv. Visualization:

A Scatter Plot was used, where each dot represents a library. This helped identify outliers and patterns in resource distribution.



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Sum of Total Library Visits
Grand Total
Total Library Visits vs Fiscal Years
HOME PAGE
Total
County
Fairfield
Hartford
Litchfield
Middlesex
New Haven
New London
Tolland
Windham
Fiscal Date
All Periods
Objective 1
Objective 2
Objective 3
Objective 4
Objective 5
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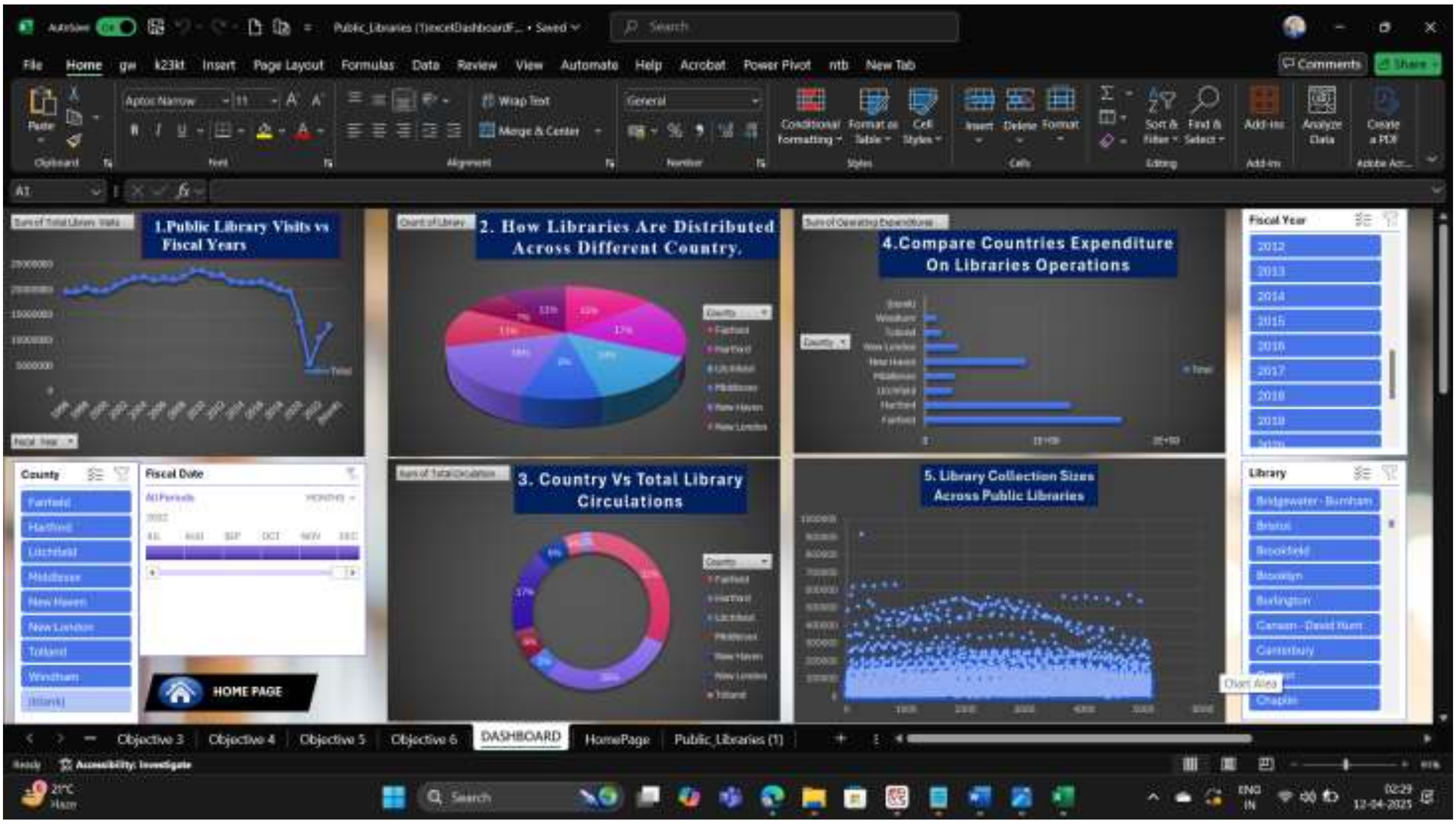
13-04-2025



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Power Pivot
New Tab
1.Public Library Visits vs Fiscal Years
2. How Libraries Are Distributed Across Different Country.
4.Compare Countries Expenditure On Libraries Operations
3. Country Vs Total Library Circulations
5. Library Collection Sizes Across Public Libraries
Fiscal Year
2012
2013
2014
2015
2016
2017
2018
2019
County
Fairfield
Hartford
Litchfield
Middlesex
New Haven
New London
Tolland
Windham
(blank)
Fiscal Date
HOME PAGE
Library
Bridgewater - Burnham
Bristol
Brookfield
Brooklyn
Burlington
Canaan - David Hunt
Canterbury
Chaplin
Objective 3
Objective 4
Objective 5
Objective 6
DASHBOARD
HomePage
Public_Libraries (1)

HOME PAGE
TOPIC . PUBLIC LIBRARIES
INT 217-
INTRODUCTION TO DATA MANAGEMENT
DASHBOARD
DATASET
SCATTER PLOT
WEBSITE LINK
DOUGHNUT
BAR GRAPH
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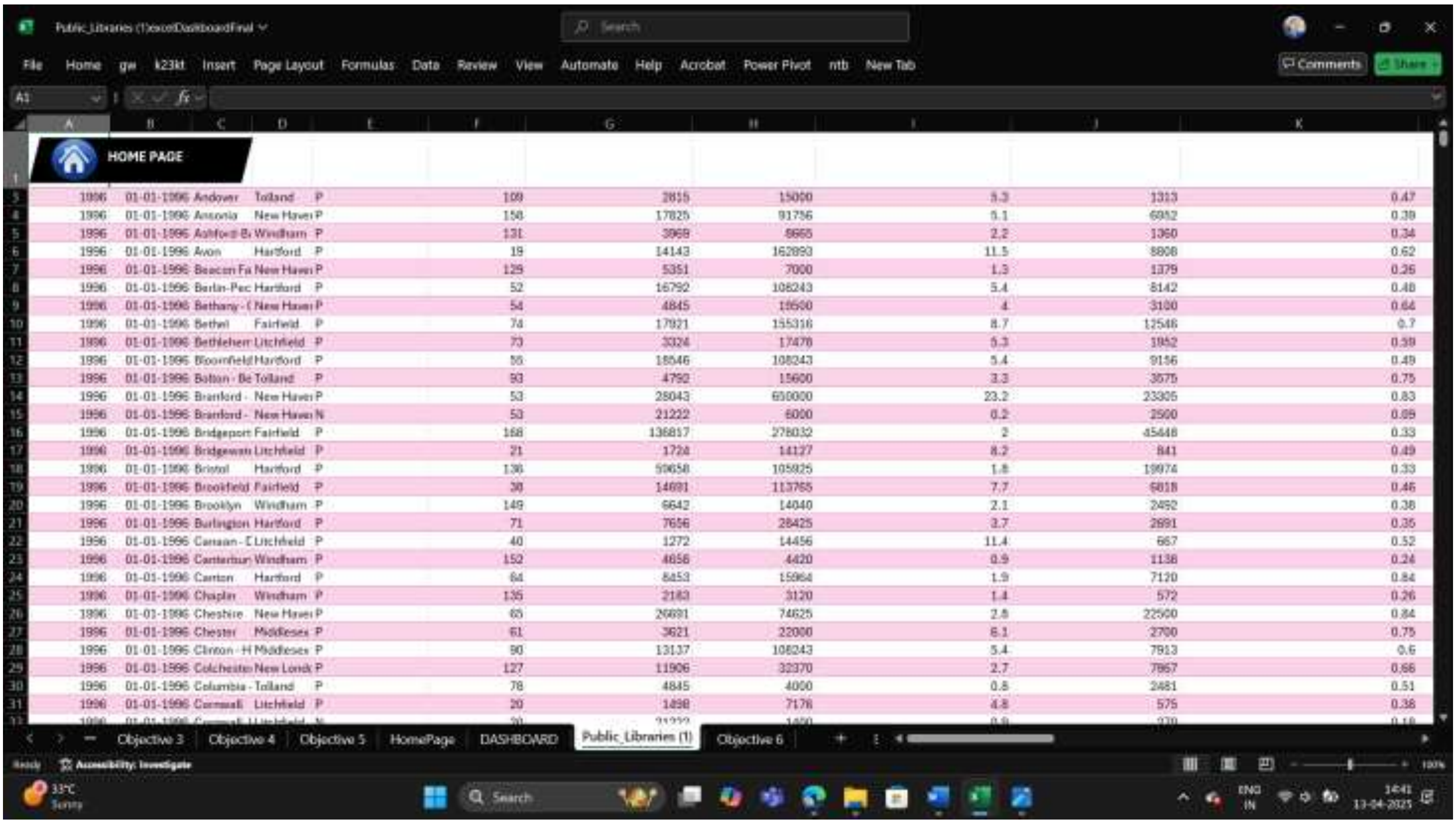
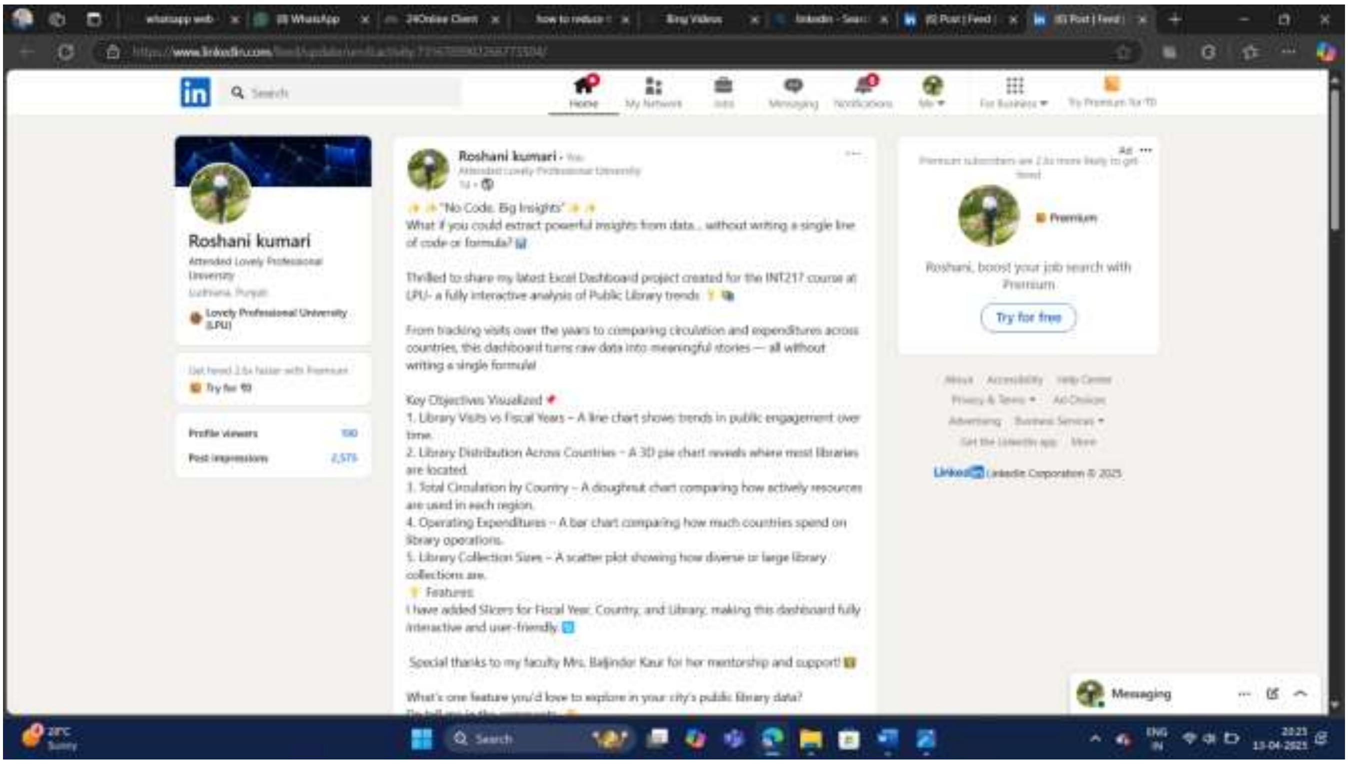
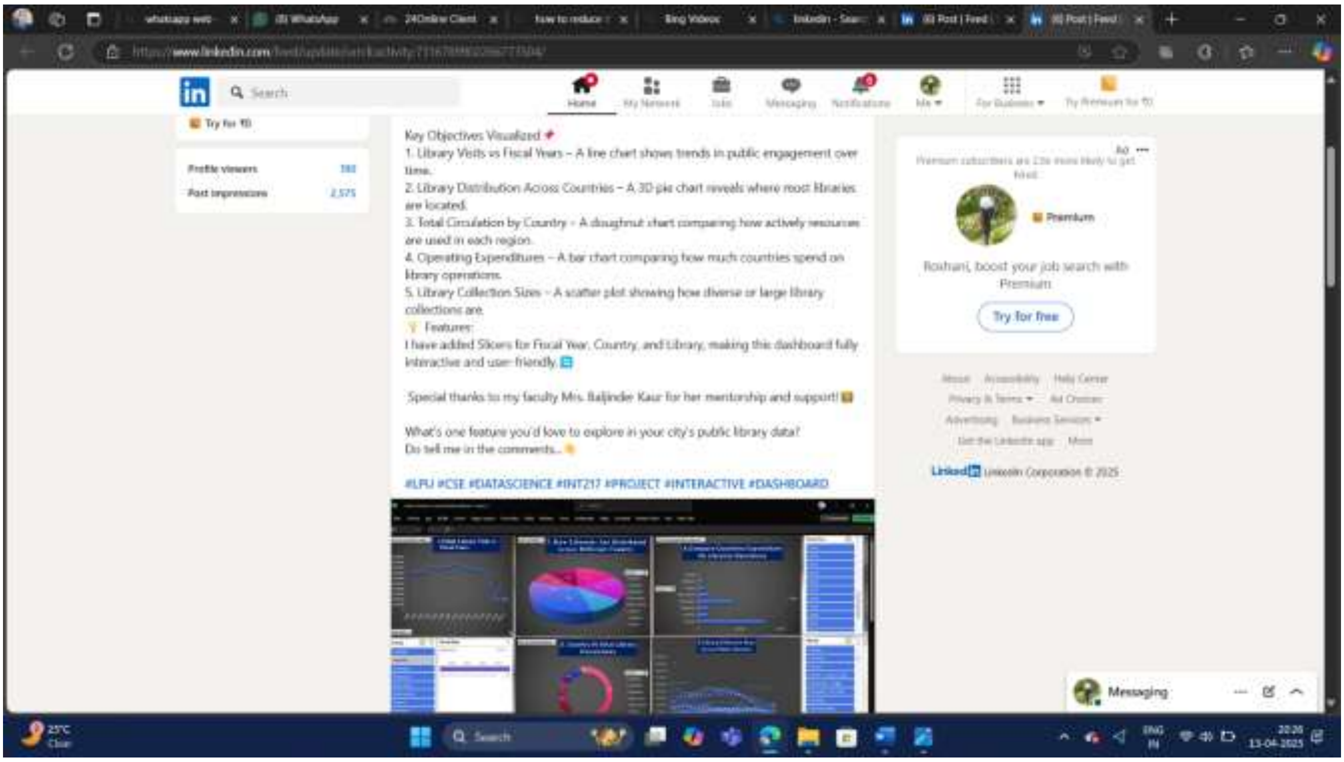


	A	B	C	D	E	F	G	H	I	J	K
3	1996	01-01-1996	Andover	Tolland	P	109	2815	15000	5.3	1313	0.47
4	1996	01-01-1996	Ansonia	New Haver	P	158	17825	91756	5.1	6952	0.39
5	1996	01-01-1996	Ashford-Bi	Windham	P	131	3969	8665	2.2	1360	0.34
6	1996	01-01-1996	Avon	Hartford	P	19	14143	162893	11.5	8808	0.62
7	1996	01-01-1996	Beacon Fa	New Haven	P	129	5351	7000	1.3	1379	0.26
8	1996	01-01-1996	Berlin-Pec	Hartford	P	52	16792	108243	5.4	8142	0.48
9	1996	01-01-1996	Bethany - (	New Haven	P	54	4845	19500	4	3100	0.64
10	1996	01-01-1996	Bethel	Fairfield	P	74	17921	155316	8.7	12546	0.7
11	1996	01-01-1996	Bethlehem	Litchfield	P	73	3324	17478	5.3	1952	0.59
12	1996	01-01-1996	Bloomfield	Hartford	P	55	18546	108243	5.4	9156	0.49
13	1996	01-01-1996	Bolton - Be	Tolland	P	93	4792	15600	3.3	3675	0.75
14	1996	01-01-1996	Branford -	New Haver	P	53	28043	650000	23.2	23305	0.83
15	1996	01-01-1996	Branford -	New Haven	N	53	21222	6000	0.2	2500	0.09
16	1996	01-01-1996	Bridgeport	Fairfield	P	168	136817	278032	2	45448	0.33
17	1996	01-01-1996	Bridgewate	Litchfield	P	21	1724	14127	8.2	841	0.49
18	1996	01-01-1996	Bristol	Hartford	P	136	59658	105925	1.8	19974	0.33
19	1996	01-01-1996	Brookfield	Fairfield	P	38	14691	113765	7.7	6818	0.46
20	1996	01-01-1996	Brooklyn	Windham	P	149	6642	14040	2.1	2492	0.38
21	1996	01-01-1996	Burlington	Hartford	P	71	7656	28425	3.7	2691	0.35
22	1996	01-01-1996	Canaan - E	Litchfield	P	40	1272	14456	11.4	667	0.52
23	1996	01-01-1996	Canterbur	Windham	P	152	4656	4420	0.9	1138	0.24
24	1996	01-01-1996	Canton	Hartford	P	64	8453	15964	1.9	7120	0.84
25	1996	01-01-1996	Chaplin	Windham	P	135	2183	3120	1.4	572	0.26
26	1996	01-01-1996	Cheshire	New Haver	P	65	26691	74625	2.8	22500	0.84
27	1996	01-01-1996	Chester	Middlesex	P	61	3621	22000	6.1	2700	0.75
28	1996	01-01-1996	Clinton - H	Middlesex	P	90	13137	108243	5.4	7913	0.6
29	1996	01-01-1996	Colcheste	New Lond	P	127	11906	32370	2.7	7967	0.66
30	1996	01-01-1996	Columbia -	Tolland	P	78	4845	4000	0.8	2481	0.51
31	1996	01-01-1996	Cornwall	Litchfield	P	20	1498	7176	4.8	575	0.38


Roshani kumari
Attended Lovely Professional University
Lovely Professional University (LPU)
Profile viewers
Post impressions
Roshani kumari
"No Code. Big Insights"
What if you could extract powerful insights from data... without writing a single line of code or formula?
Thrilled to share my latest Excel Dashboard project created for the INT217 course at LPU- a fully interactive analysis of Public Library trends
From tracking visits over the years to comparing circulation and expenditures across countries, this dashboard turns raw data into meaningful stories — all without writing a single formula!
Key Objectives Visualized
1. Library Visits vs Fiscal Years – A line chart shows trends in public engagement over time.
2. Library Distribution Across Countries – A 3D pie chart reveals where most libraries are located.
3. Total Circulation by Country – A doughnut chart comparing how actively resources are used in each region.
4. Operating Expenditures – A bar chart comparing how much countries spend on library operations.
5. Library Collection Sizes – A scatter plot showing how diverse or large library collections are.
Features
I have added Slicers for Fiscal Year, Country, and Library, making this dashboard fully interactive and user-friendly.
Special thanks to my faculty Mrs. Baljinder Kaur for her mentorship and support!
What's one feature you'd love to explore in your city's public library data?
Premium
Roshani, boost your job search with Premium
Try for free
Messaging



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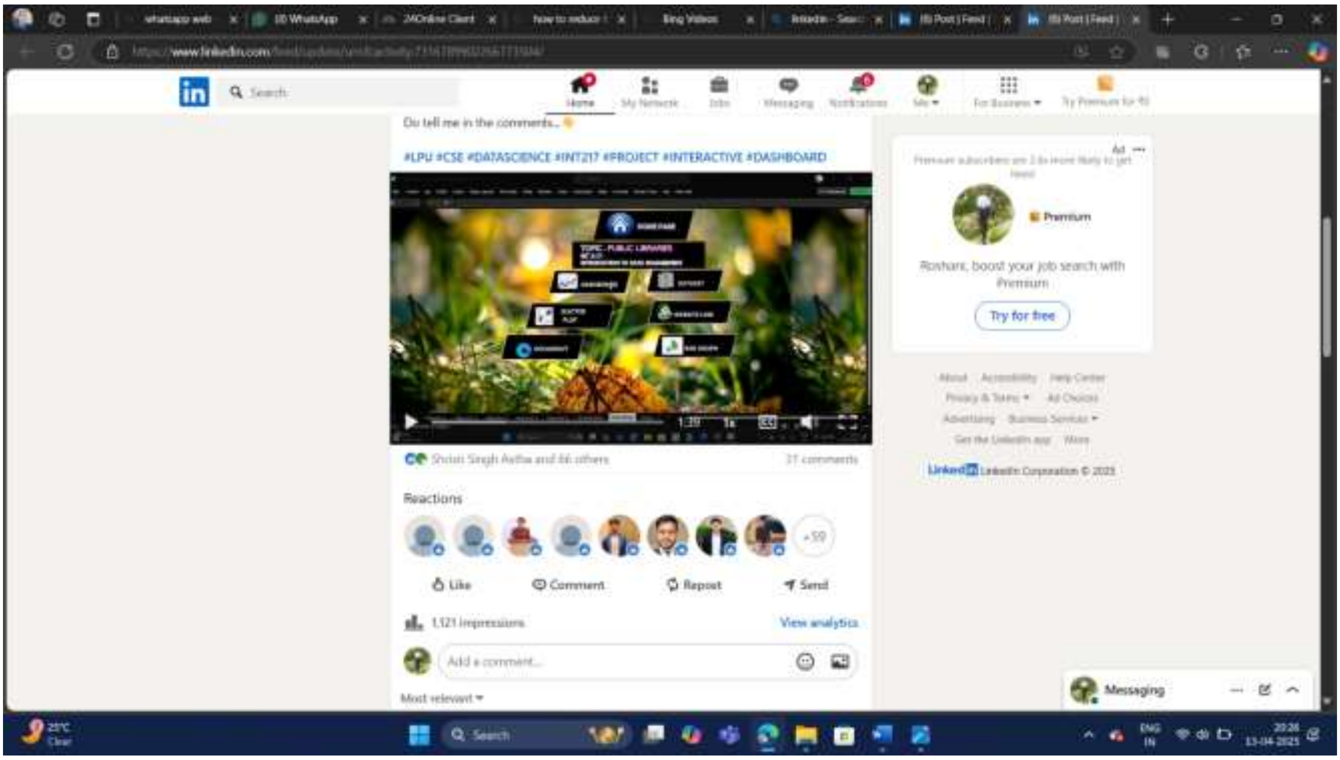
What's one feature you'd love to explore in your city's public library data?
Do tell me in the comments...
#LPU #CSE #DATASCIENCE #INT217 #PROJECT #INTERACTIVE #DASHBOARD
Premium
Try for free
Messaging



Do tell me in the comments...
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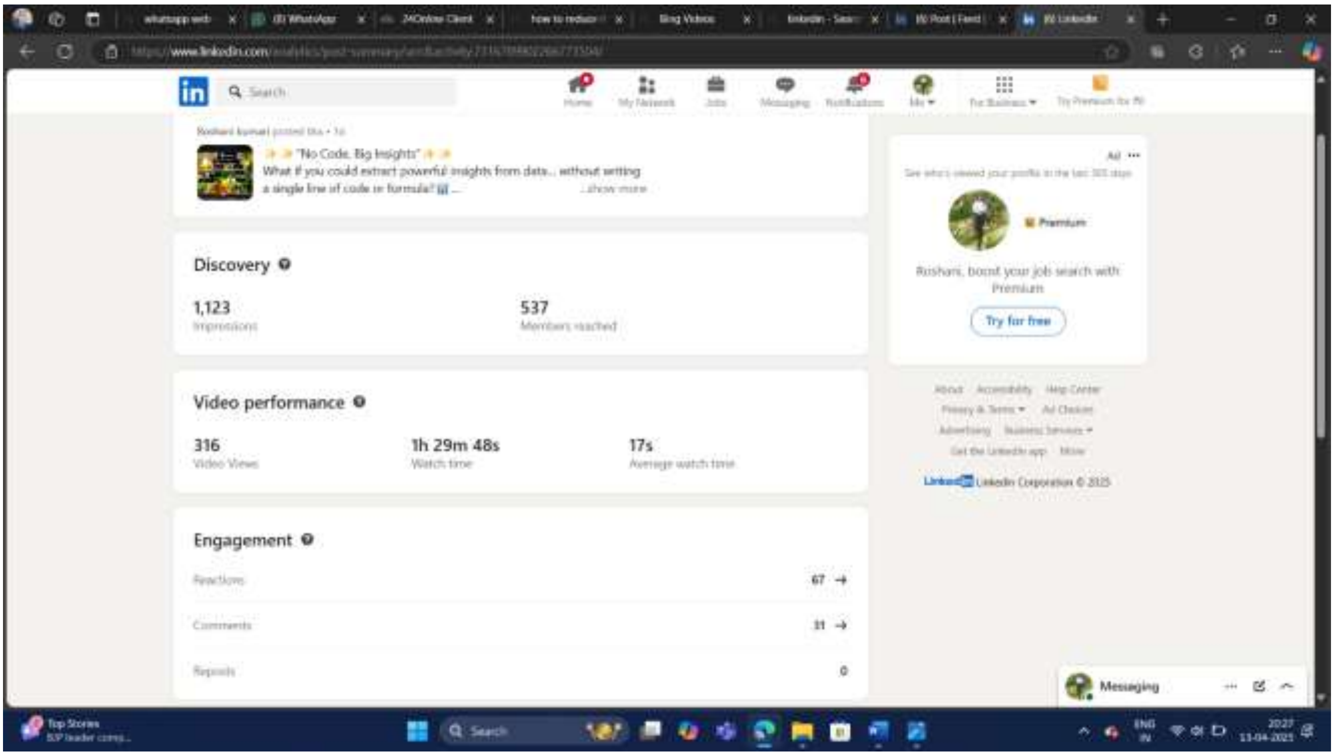
"No Code, Big Insights"
What if you could extract powerful insights from data... without writing a single line of code or formula?
Discovery
1,123
Impressions
537
Members reached
Video performance
316
Video Views
1h 29m 48s
Watch time
17s
Average watch time
Engagement
Reactions
67
Comments
31
Reposts
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Try for free
Messaging

5. Conclusion

The Excel Dashboard project on Public Libraries provided a comprehensive platform to explore and analyze library-related data across various counties and fiscal years. By leveraging pivot tables, slicers, and a wide range of charts—without the use of formulas—the project effectively transformed raw data into meaningful insights.

Through the first objective, we observed dynamic trends in public library visits across fiscal years, which helped identify periods of high and low public engagement. The second objective presented a visual understanding of how library infrastructure is geographically distributed, revealing countywise variations in the number and types of libraries. Moving forward, the third objective focused on library circulations, helping us assess the extent of resource utilization in each region. The fourth analysis gave a financial lens by comparing operational expenditures across counties, revealing the level of investment in public library services. Lastly, examining the collection sizes in the fifth objective allowed us to understand how resources are distributed and where potential disparities exist.

Overall, this project not only developed a deeper understanding of public library systems but also enhanced practical skills in data visualization, storytelling through dashboards, and making datadriven observations. The dashboard format made the information accessible, interactive, and easy to interpret for academic and administrative audiences alike.

This project reaffirms how powerful Excel can be as a data analysis tool, especially when used strategically with pivot charts, slicers, and visual formats tailored to different objectives.

6. Future Scope

While the current Excel Dashboard successfully highlights key trends and insights related to public libraries, there remains significant potential to enhance the analysis and broaden the impact of this project in the future. As data analytics continues to evolve, several areas could be explored to add more depth, interactivity, and value to the existing dashboard.

Firstly, integrating additional datasets—such as demographic data (age groups, literacy rates, internet access), educational statistics, or community engagement metrics—could help correlate library usage with external factors, offering richer, multi-dimensional insights. This would allow for a more detailed understanding of why certain counties experience higher or lower visit counts or circulation numbers.

Secondly, the dashboard could be expanded with advanced Excel features like Power Query and Power Pivot to manage larger datasets and create more dynamic, relational models. This would help overcome some limitations of standard pivot tables and enable real-time updates as new data becomes available.

Additionally, incorporating time-series forecasting or trendline analysis could help predict future library usage patterns, budgeting needs, or resource demands. These predictive insights could prove valuable to government agencies or library management teams when planning policies or allocating resources.

There is also potential to migrate the dashboard to more powerful data visualization platforms like Power BI or Tableau. Doing so would introduce features such as automated filtering, drill-down capabilities, and live data connections, making the dashboard even more interactive and user-friendly for stakeholders.

Finally, user feedback could be integrated into future versions of the dashboard, enabling personalized views and report generation based on specific queries or user types (e.g., administrators, students, researchers).

In conclusion, the foundation built through this Excel project can evolve into a more advanced, scalable, and insightful analytical tool. With further development, it can serve not just as an academic submission, but as a meaningful asset in understanding and improving the operations of public libraries.

7. References

[1] U.S. General Services Administration, “Public Libraries Dataset,” Data.gov. [Online]. Available: https://data.gov. [Accessed: Apr. 12, 2025].

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